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Primary Knee

Using Unsupervised Machine Learning to Predict Quality of Life After Total Knee Arthroplasty

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ABSTRACT

Background: Patient-reported outcome measures (PROMs) are an important metric to assess total knee arthroplasty (TKA) patients. The purpose of this study was to use a machine learning (ML) algorithm to identify patient features that impact PROMs after TKA.

Methods: Data from 636 TKA patients enrolled in our patient database between 2018 and 2022, were retrospectively reviewed. Their mean age was 68 years (range, 39 to 92), 56.7% women, and mean body mass index of 31.17 (range, 16 to 58). Patient demographics and the Functional Comorbidity Index were collected alongside Patient-Reported Outcome Measures Information System Global Health v1.2 (PROMIS GH-P) physical component scores preoperatively, at 3 months, and 1 year after TKA. An unsupervised ML algorithm (spectral clustering) was used to identify patient features impacting PROMIS GH-P scores at the various time points.

Results: The algorithm identified 5 patient clusters that varied by demographics, comorbidities, and pain scores. Each cluster was associated with predictable trends in PROMIS GH-P scores across the time points. Notably, patients who had the worst preoperative PROMIS GH-P scores (cluster 5) had the most improvement after TKA, whereas patients who had higher global health rating preoperatively had more modest improvement (clusters 1, 2, and 3). Two out of Five patient clusters (cluster 4 and 5) showed improvement in PROMIS GH-P scores that met a minimally clinically important difference at 1-year postoperative.

Conclusions: The unsupervised ML algorithm identified patient clusters that had predictable changes in PROMs after TKA. It is a positive step toward providing precision medical care for each of our arthroplasty patients.

Healthcare has entered an era of precision medicine, defined as a way to provide individualized patient care utilizing the power of data [1]. This is possible because the application of artificial intelligence within clinical practice continues to mature and become more commonplace [2,3]. Machine learning (ML) is a subfield of artificial intelligence and it includes algorithms that can "learn" patterns from data for prediction or classification. The field of total joint arthroplasty is full of potential to use ML algorithms because of the access to rich and relatively large data sets [4].

Precision surgical care is especially relevant to the many patients suffering from knee osteoarthritis who undergo total knee arthroplasty (TKA); indeed there were over 55,000 knee arthroplasties performed in Canada in 2021 [5]. The goal of a TKA is to reduce pain and enhance patients' overall quality of life, yet consistently 12 to 30% of TKA patients are not satisfied with their outcome despite modern surgical advances [5–8]. Machine learning (ML) algorithms can identify patient features that affect

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outcomes after arthroplasty opening the door to more individually targeted interventions [9,10].

Delivering the highest quality of care remains a priority not just for surgeons, but also for funding agencies (ie, government or private carriers). A key motivator behind offering a bundled payment model for joint arthroplasty is the high-quality, efficient, and coordinated surgical care for patients. Patient-reported outcome measures (PROMs) represent a key performance indicator for these bundled payments.

The purpose of this study was to apply a ML algorithm to identify and describe these important patient features that impact quality of life outcomes according to the Patient-Reported Outcomes Measurement Information System (PROMIS) Global Health v1.2 scale.

Methods

Ethics review board approval was obtained to retrospectively review patients who had total knee arthroplasties enrolled in the prospective patient database collected by the Division of Orthopaedics at the Ottawa Hospital. This database was created in 2018 to inform continuous quality improvement in our center. Patients are asked to provide their consent to have their data included in this database when they are booked for elective orthopaedic surgical procedures.

Data Collection and Extraction

Data were abstracted from patients who underwent TKA (unilateral) between October 1, 2018, and March 5, 2022 and were enrolled in the Division of Orthopeedic Surgery continuous quality improvement database as above. The following patient data were abstracted: age at the time of surgery, gender, diagnosis, procedure type, procedure side, rate of reported comorbidity from the items outlined in the Functional Comorbidity Index (self-administered index of comorbid diseases) [11], and Patient-Reported Outcome Measures Information System Global Health v1.2 (PROMIS GH-P) physical component scores [12]. Patient-Reported Outcomes Measurement Information System (PROMIS) Global Health is a self-rated tool that evaluates physical and mental health and was collected preoperatively, 3 months, and 1 year following surgery. Scores obtained from this scale were used as the predictor variable representing patients' quality of life. The minimally clinically important difference (MCID) reference values were determined according to prior work validating the responsiveness of this scale in knee arthroplasty [13]. Also incorporated in our analysis was data on body mass index (BMI) retrieved from the hospital data warehouse. There were 1,737 patients who had undergone a TKA available for analysis, 36% of patients had complete data and were included in the analysis.

Exclusion Criteria

Patients less than 18 years of age, with single-setting bilateral TKA and who did not have complete data at the preoperative, 3-months postoperative, or 1 year postoperative time intervals were excluded.

This allowed 636 patients who had a unilateral knee arthroplasty to be included in the dataset for the application of ML algorithm, mean age 68 years (range, 39 to 92) (standard deviation, 8.1), 56.7% women, mean BMI 31.48 (range, 16 to 58). There were no statistical differences found in age (mean age 67.87, $P = .57$), gender distribution (58.6% women, $P = .46$) or BMI (mean 31.67, $P = .62$) between the patients included and those who have incomplete data who were excluded.

Application of the ML Algorithm

Considering that our goal was to better understand possible patterns in data to identify patient features that impact global health outcomes, we used an unsupervised ML algorithm. Unsupervised ML algorithms use unlabelled data and are allowed to act on that data without constraint, or "supervision". An unsupervised ML algorithm can find underlying structures in a dataset and then group data according to similarities. More specifically, this study used a spectral clustering (SC) algorithm that belongs to a class of unsupervised ML methods. Spectral clustering (SC) is a clustering algorithm rooted in graph theory where individual data instances (patients) are considered as nodes in a graph and therefore data clustering becomes a graph partitioning problem. The SC algorithm partitioned data following 3 distinct steps: first, a similarity graph for all data points was constructed; secondly, data points were embedded in more evident clusters using the eigenvectors of the graph. In the third step, classical clustering algorithm, such as k-means partitioned the embedding [14]. The clusters are illustrated using t-SNE (T-distributed Stochastic Neighbor Embedding method (https://scikit-learn.org/stable/modules/generated/sklearn.manifold.TSNE.html). In our study, we used the Scikit-learn package that was developed in Python (https://scikit-learn.org/stable/index.html).

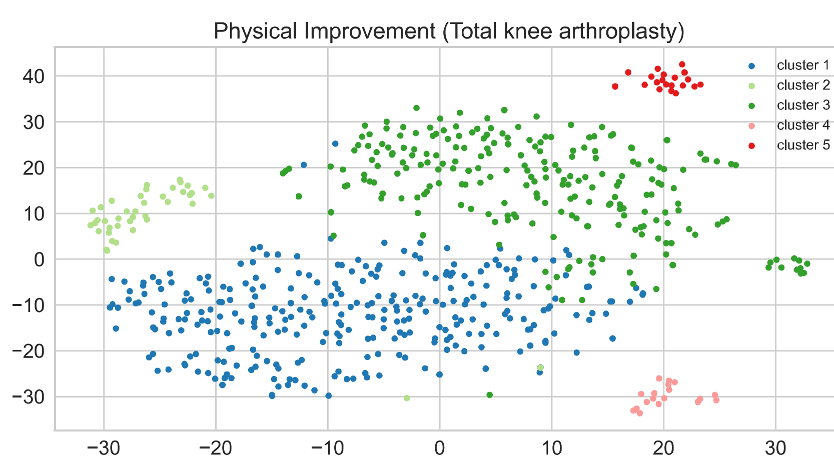

Fig. 1. Five patient clusters.

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Table 1
Summative Description of 5 Clusters.

Criteria	Cluster 1	Cluster 2	Cluster 3	Cluster 4	Cluster 5
N	306	41	245	21	23
Mean Age	68	69	67	67	61
Men (%)	42	51	41	62	48
Women (%)	58	49	59	38	52
Respiratory (%)	10.5	4.9	20.8	23.8	13.0
Cardiovascular (%)	24.5	14.6	34.3	14.3	21.7
Mental (%)	17.0	14.6	30.2	33.3	21.7
Arthritis (%)	99.0	0	97.1	85.7	82.6
Osteoporosis (%)	23.9	19.5	29.4	19.0	17.4
Neurological Disease (%)	0	0	4.5	0	0
Upper Gastrointestinal Disease (%)	25.5	12.2	31.0	0	26.1
Body Mass Index	30.2	29.3	32.8	29.6	30.9
Average Preoperative Pain Score	4.8	4.7	6.5	6.6	8.0

All the values of outcomes in data set were standardized to have mean of 0 and standard deviation of 1.

Results

Identifying Patient Features

Application of the SC algorithm on the dataset produced 5 patient clusters as illustrated in Figure 1. These 5 clusters are described in Table 1 according to age, gender, BMI and the rate of reported comorbidities from the Functional Comorbidity Index. Some notable findings include that clusters are distinguishable by age, gender, and BMI features. Clusters 1 and 3 are majority women patients (58 and 59% respectively), cluster 4 is majority men (62%), while clusters 2 and 5 are more evenly split. Cluster 5 is the youngest grouping of patients (average of 60 years). Cluster 3 includes patients having the highest average BMI (32.8). The average pain scores are noted for each cluster: cluster 1 and 2 have the lowest average preoperative pain scores (4.8 out of 10 and 4.7 out of 10, respectively), cluster 3 and 4 patients have higher preoperative scores (6.5 out of 10 and 6.6 out of 10), while cluster 5 patients have the highest preoperative score (8/10).

Patients self-report a high rate of various comorbidities across all the clusters. Using cluster 1 as an example, of the patients that were identified by the algorithm into this cluster about ¼ of them also suffer from cardiovascular disease (24.5%), osteoporosis (23.8%) and upper gastrointestinal disease (25%). They have comparatively lower rates of respiratory (10%), neurological (0%), or mental health disorders (16%) than other groups. The full side-by-side comparison is in Table 1. Interestingly, no patients in cluster 2 report arthritis in any other joint indicating that these patients have monoarticular disease of their knee, whereas nearly 100% of patients in Clusters 1 and 3 have reported being affected by osteoarthritis in other joints, and Cluster 4 and 5 have similarly high rates with 85.7% and 82.6% of patients in these groups self-reporting comorbid arthritis of another joint.

Understanding Patient Reported Outcomes by Patient Cluster

Table 2 describes the clusters' PROMIS GH-P scores at the initial, 3-month, and 1-year postoperative time points. The percentage of patients in each cluster who reached the MCID for clinical improvement is also shown in Table 2. Figure 2 graphically illustrates the temporal relationship of the PROMIS GH-P scores at the preoperative/3-month postoperative and 1-year postoperative time points, with a density function to better assess relationships over time. Patients in cluster 1 show the least improvement of all the groups. They have high starting PROMIS GH-P scores (13.6). Postoperative scores (14.0 at 3 months and 14.96 at 1 year) after TKA, while slightly improved, do not meet MCID for this scale in any patient in the group. Average cluster 2 preoperative PROMIS GH-P scores are highest of all the groups (13.7) but show improvement in their scores to an average of 15.3 at 3 months and 16.1 at 1 year, with 34% of patients meeting the MCID for improvement in the PROMIS GH-P score at 3 months and this is maintained at 1 year. No patients in this cluster met the MCID at 1 year. Similar outcomes can be identified for cluster 3. Cluster 3 average PROMIS GH-P preoperative scores are 10.7, with improvement to an average of 14.5 at 3 months and 13.8 at 1 year. While 87% of the patients in this cluster report improved PROMIS GH-P scores at 3 months, scores slightly decrease between the 3-month and 1-year time point. Cluster 3 patients have the lowest PROMIS GH-P scores after 1 year. Cluster 4 patients have an average preoperative PROMIS GH-P score of 11.6 and very slightly decreased scores at 3 months (11.5) before showing improvement in the scores at 1 year (17.3). Cluster 4 patients are the only group to have a decrease in scores within the first 3 months of TKA. Twenty-three percent of cluster 4 patients meet the MCID between their preoperative and 1-year evaluations

Table 2
Patient-Reported Outcome Measures: By Cluster.

Outcome Measures	Cluster 1	Cluster 2	Cluster 3	Cluster 4	Cluster 5
PROMIS Global Physical Preop	13.7	13.8	10.7	11.6	8.8
PROMIS Global Physical 3 mo	14.0	15.3	14.5	11.5	15.8
PROMIS Global Physical 1 y	15.0	16.2	13.8	17.3	18.0
PROMIS Improvement 3 mo to pre	0.37	1.56	3.80	−0.10	7.04
PROMIS Improvement 1 y to pre	1.31	2.44	3.04	5.71	9.13
PROMIS Improvement 1 y to 3 mo	0.94	0.88	−0.76	5.81	2.09
MCID_pre_3 m (≥2.5) (%)	0.6	34	87	10	100
MCID_pre_1 y (≥7.5) (%)	0	0	0	23	100
MCID_3m_1 y (≥5) (%)	0	0	0	100	0

PROMIS, Patient-Reported Outcomes Measurement Information System; MCID, minimally clinically important difference.

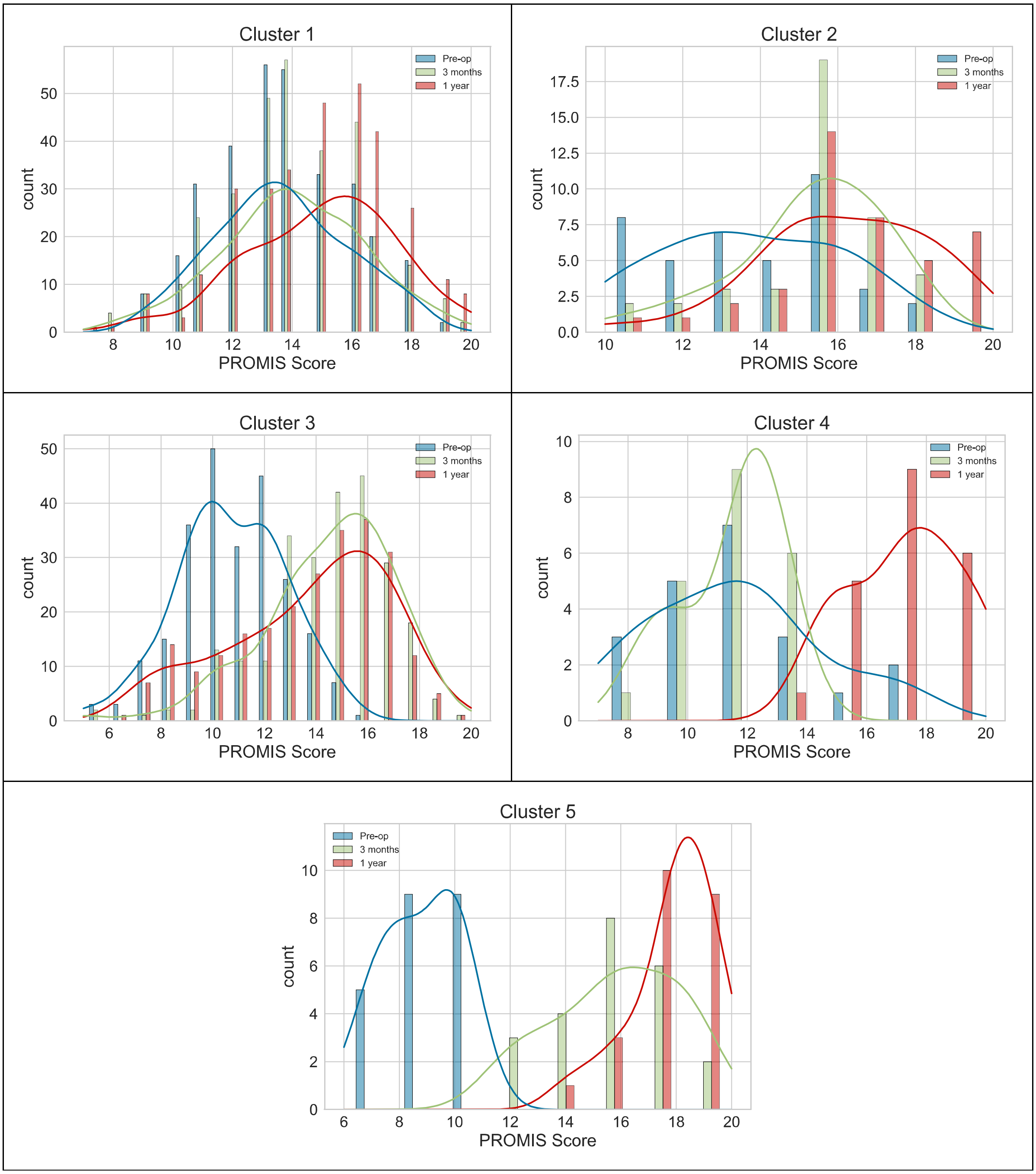

Fig. 2. Temporal changes of the Patient-Reported Outcomes Measurement Information System (PROMIS) scores.

and all of them meet the MCID between 3 months and 1 year. Cluster 5 patients show the most improvement at 3 months and 1 year. Cluster 5 average preoperative scores are 8.8 (the lowest of the clusters), improving to 15.8 at 3 months and 17.9 at 1 year. One hundred percent of the patients in this cluster meet the MCID for improvement between their preoperative and 3-month/1-year time points.

Cluster 5 patients, with the largest improvement in PROMIS GH-P scores, show improvements early and are able to sustain those improvements. They are also the youngest and have the highest starting pain scores. Other notable relationships between the cluster features and the PROMIS GH-P scores exist for clusters 3 and 4. Cluster 3 patients have the highest self-reported rates of comorbidities in almost all categories except respiratory and mental health conditions; they also have the highest BMI, This is the only group with a decrease in their PROMIS GH-P scores between the 3-month and 1-year mark and they end up with the lowest overall PROMIS GH-P scores. Cluster 4 patients with the highest rates of respiratory and mental health conditions are the only grouping of patients that show very minimal decrease in PROMIS GH-P scores within the first 3 months after TKA, but they eventually do well with improvements in their scores at 1 year, 100% of patients showing change that meets a MCID. To better get a sense of what an average patient in each

Table 3
Centroid of Each Cluster (Exemplar).

Features	Cluster 1	Cluster 2	Cluster 3	Cluster 4	Cluster 5
Age (y)	69	69	67	67	63
Gender	Women	Men	Women	Men	Women
Body Mass Index	30.2	29.4	31.4	30.4	31.6
General health[a]	4	4	3	3	3
Physical health[a]	3	3	3	3	2
Usual activities[a]	4	4	3	4	2
Physical activities[a]	4	3	3	3	2
Fatigue[a]	4	4	3	3	3
Average Pain Scale (0 = none, 10 = severe)	5	5	7	7	8
Concurrent Arthritis	yes	no	yes	yes	yes
Other Comorbidities[b]	no	no	no	no	No
Global_Physical_Pre	14	14	11	12	9
Global_Physical_3 mo	14	16	15	12	16
Global_Physical_1 y	15	16	14	17	18
3 mo_pre	1	1	4	0	7
1 y_pre	1	3	3	6	9
1 y_3 mo	1	1	0	6	2
MCID_pre_3 mo	no	no	yes	no	yes
MCID_pre_1 y	no	no	no	no	yes
MCID_3 mo_1 y	no	no	no	yes	No

MCID, minimally clinically important difference.
[a] Likert Scale 1 = Poor, 5 = Excellent.
[b] Other comorbidities as self-reported from the Functional Comorbidity Index: osteoporosis, upper gastrointestinal, degenerative disc, neurological, respiratory, cardiovascular, or mental health diseases.

cluster would look like, the "centroid" for each cluster is described in Table 3.

Discussion

Patient-reported outcome measures (PROMs) have become an important pillar in the assessment of how we deliver care from the individual surgeon to institution and payer-led/government assessment programs as they provide a more comprehensive insight to our patients' general well-being and quality of life before, and after surgical treatment. Surgeons should expect that reliance on these metrics will continue. Total knee arthroplasty (TKA) has been projected to increase between 43 and 276%, depending on global region [15–17]. This will put immense pressure on healthcare systems and measuring patient outcomes will continue to be an important part of the multipronged approach to managing this demand [18].

To apply ML algorithms successfully, a data set needs to contain clear and well-defined features to be used by an algorithm [19]. As a first step in data exploration, an unsupervised ML algorithm can be used to search for the patterns in data and to establish best ways to describe these patterns from clinical perspective. This allows researchers and clinicians to be nimbler in further exploration of data for clinically meaningful insights. Our unsupervised algorithm has identified 5 discernable clusters of patients who have undergone unilateral TKA. We have described these clusters in the results above to gain insight into the clinical relevance of these clusters in our patient population. In general, these patient clusters are characterized by the features we would expect from medical literature that is, the predominantly women groups also having the highest rates of concurrent arthritis and osteoporosis, but that in general, the patient population is multimorbid. Identified clusters are associated with expected trends of PROMIS GH-P scores predicting the impact of a TKA on patients' overall general health. The most improvement was seen in the cluster of patients who have the youngest age and worst preoperative starting pain scores indicating that the TKA had the highest positive effect on their overall global health. The cluster with highest-reported mental health concerns had the least early improvement but did well at 1 year. The cluster of patients who have the highest rates of comorbidities across many of the conditions and the highest BMI had the lowest PROMIS GH-P scores at 1 year.

Other published work demonstrates that ML algorithms are used for predicting specific surgical outcomes such as cost, length of stay, risk of complications, or surgery length [10,20]. Machine learning (ML) algorithms are also being utilized to identify patient factors that predict satisfaction and patient-reported Hip Disability and Osteoarthritis Outcome scores [21,22]. We are not aware of another study such as ours that identifies patient factors that impact PROMs. Data-driven methods such as the ML algorithms will play an important role in helping to establish why a residual number of patients are unsatisfied after TKA. Since funding models and surgeons are increasingly relying on PROMs to track patient outcomes after arthroplasty, our study points in a right direction.

Patient-reported outcome measures (PROMs) have limitations. While the PROMIS GH-P score had been shown to be as responsive in knee arthroplasty as the knee-specific Knee Injury and Osteoarthritis Outcome Score [13]; our results have shown it has a ceiling effect, whereby if patients are relatively well before their surgery, the ability to nuance measurable change after surgery is limited. Thus, as one might expect, our patient group with the worst preoperative scores had the biggest relative improvements in their quality of life as measured by PROMIS GH-P postoperatively. Patients who are relatively well, or who's quality of life is otherwise affected by their comorbidities do not show as profound a change in their PROM. One plausible explanation being that PROMIS GH-P scale is not sensitive enough for measuring nuanced outcomes of the TKA. Other study limitations include the fact that unsupervised ML algorithms, while helpful in identifying patterns in data and impactful features, do not allow to establish causality, but rather points toward an association. Additionally, if identified clusters are going to be used for patients' categorization before the surgery, it might be difficult to assign a patient to a most appropriate cluster at this time point based solely on demographics and preoperative data. Also, the results are only as good as data. We need to acknowledge that our study, while generalizable, was conducted using a single center's data and the number of patients who had PROMs reported at required time points was relatively small. A minority of patients who were included in the retrospective review of the database had complete demographic and scoring information for inclusion in the algorithm, which could lead to an unintended sampling bias. As our study was not aimed at measuring change or impact of an intervention, rather completed to better our understanding of our patient population, the selection of the study population for the algorithm was determined by the availability of complete data. Our results are comparable to response rates of large cohorts for PROMs [23,24]. If the authors were to use a ML algorithm to compare groups or interventions in another study, it would be prudent to fill in the gaps of missing data to best negate the impact of unintended sampling bias.

This study is another step in understanding the patient features that affect quality of life after TKA. As we strive to provide precision medicine to the patients at our center, we will continue to collect these important patient features in our database. The features defined by the ML algorithm can be translated to larger data sets both at other institutions and/or in follow-up studies on the longitudinally collected patients within this database. Our results can be used as the control to assess changes we make to the delivery of care or the effect of a patient intervention on the TKA outcome.

Conclusions

Patient-reported outcomes will continue to be central to evaluating outcomes of the TKA at our center and others across the globe. Our study has applied an *unsupervised* ML algorithm to our dataset and has identified 5 patient clusters within our patient population that had predictable PROMIS GH-P scores. We then described the features of these groups to better understand the clinical relevance. Now, if we were to apply an intervention to our population, we could harness the powerful methodology of ML in a *supervised* manner to achieve a deeper analysis with the ML algorithm than classical statistical models have been able to achieve. While we cannot yet predict specific patient outcomes based on this descriptive work, repeated application of a supervised ML algorithm to our dataset or other combined datasets has the potential to use these features in a predictive fashion. This study is one step toward provision of precision medical care to each of our patients.

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